

Enrolment No:.....

Name of Student.....



BENNETT
UNIVERSITY
THE TIMES GROUP

NAAC
GRADE A+
ACCREDITED UNIVERSITY

End Semester Examinations, May 2026

Even Semester 2025-26

B.Tech.(CSE)/B.Tech. (EP) – IVth Semester

CSET207 - COMPUTER NETWORKS

Time: 2hrs.

Max Marks: 40

Note: *Student must attempt 5 Questions in total. First question is compulsory and student must attempt question(s) from each section. Marks are indicated against each question.*

PART-A

Q.1 Attempt any 6.

- (a) Discuss the problems that would arise if there were no layered architecture in computer networks? [CO-1] [L-2] 2
- (b) Explain Packet Switching. How is it different from Circuit Switching? [CO-2] [L-2] 2
- (c) Discuss the impact of central hub/switch failure in a Star topology? How does this compare to Bus and Ring topologies? [CO-2] [L-2] 2
- (d) Analyze why variable-length framing is considered more complex yet more reliable than fixed-length framing in data link layer communication. Justify your answer with appropriate reasoning. [CO-2] [L-2] 2
- (e) Explain why the two IP addresses (network and broadcast) not usable for hosts in every subnet? [CO-2] [L-2] 2
- (f) Analyse and justify why sockets are essential for enabling process-to-process communication instead of merely supporting host-to-host communication in computer networks. [CO-2] [L-3] 2
- (g) Consider the following 32-bit binary sequence from a TCP header:
11000000 00010110 00000000 01010000
Identify the Source Port number and Destination Port number [CO-2] [L-3] 2
- (h) Analyse the role of trunk ports in VLAN-based network design and evaluate the impact on network communication if trunking is not enabled between switches. [CO-3] [L-2] 2

2*6=12 Marks

PART-B

Attempt any two questions.

- Q.2 A network administrator has the following networks: 172.16.0.0/24, 172.16.1.0/24, 172.16.2.0/24, 172.16.3.0/24, 172.16.4.0/24, 172.16.5.0/24, 172.16.6.0/24, 172.16.7.0/24. The administrator summarizes the routes through a particular port of the router for lesser advertisement overheads.
- Write the summarized address in CIDR notation and Decimal subnet mask.
 - Specify the range of IP addresses covered by the summarized route.
- iii. Explain how route summarization helps in reducing routing table size. [CO-2] [L-3] 6
- Q.3 A router needs to be configured using **OSPF** with the following requirements:
- Process ID = 20
- Router ID = 2.2.2.2
- Networks to advertise:
- 172.16.1.0/24 → Area 0
- 172.16.2.0/24 → Area 0
- 172.16.3.0/24 → Area 1
- 192.168.100.0/24 → Area 2
- Write the complete configuration commands
 - Explain necessity to assign a router ID in OSPF with the significance of using multiple areas.
- iii. Explain what will happen if Area 0 is missing in the configuration. [CO-2] [L-3] 6
- Q.4 Design a codeword that guarantees detection of two-bit errors, given a two-bit Data word. [CO-2] [L-3] 6
- 6*2=12 Marks**

PART-C

Questions with an internal choice.

- Q.5 (a) Detail the working of Dynamic Host Configuration Protocol using an illustrated transactional diagram between Client and DHCP server. (Diagram Only). [CO-3] [L-2] 8
- OR**
- Q.5 (b) Discuss Domain Name Space. Shed some light on Iterative and Recursive DNS Resolution. [CO-3] [L-2] 8
- Q.6 (a) If one layer in the network stack fails or behaves inefficiently (e.g., congestion at transport layer or errors at data link layer), how does it affect overall network performance? Explain the interdependence of layers with examples. [CO-3] [L-2] 8
- OR**
- Q.6 (b) Explain the structure of the IPv4 header and discuss how its fields support reliable and efficient packet delivery across networks. Highlight the role of fragmentation and TTL. [CO-3] [L-2] 8
- 8*2=16 Marks**
